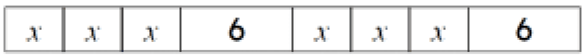
Name: ______________________________ Starter quiz

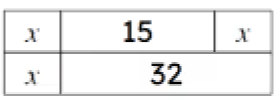
Equality in an equation

1 Which expression is represented by this bar model? Tick **2** correct answers



- ☐ $3x + 6$
- ☐ $6x + 6$
- ☐ $6x + 12$
- ☐ $2(3x + 6)$
- ☐ $3(x + 6)$

2 Select the bar model that correctly represents the equation $3x + 15 = x + 32$. Tick **1** correct answer



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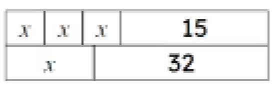

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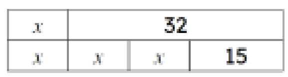

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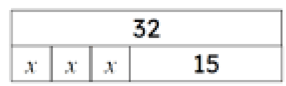

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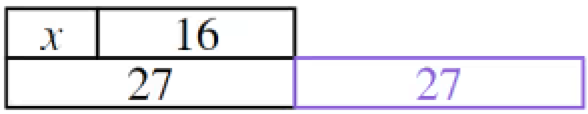
3 Given that $2.4 + 3.7 = 6.1$, which of these are also true? Tick **2** correct answers

- ☐ $2.4 + 6.1 = 3.7$
- ☐ $6.1 - 2.4 = 3.7$
- ☐ $3.7 - 2.4 = 6.1$
- ☐ $2.4 = 6.1 - 3.7$

4 This bar model represents the initial equation $x + 16 = 27$. What needs to be done to the top bar to maintain equality? Tick **2** correct answers

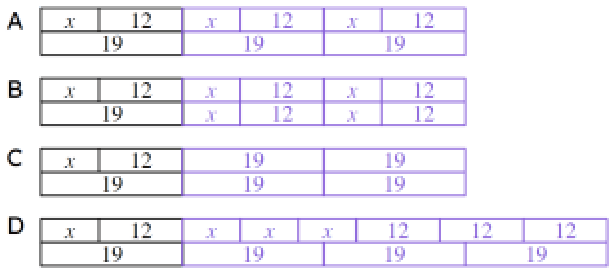


- ☐ $+ 2$
- ☐ $+ 16$
- ☐ $+ (x + 16)$
- ☐ $+ 27$
- ☐ $+ x$

5 This bar model represents the initial equation $2x + 5 = 9$. What needs to be done to the top bar to maintain equality? Tick **2** correct answers

- ☐ $+ 9$
- ☐ $+ (2x + 10)$
- ☐ $+ (4x + 10)$
- ☐ $\times 2$
- ☐ $\times 3$

6 These 4 bar models represent operations performed on the equation $x + 12 = 19$. Match the bar models labelled A, B, C, D with the equivalent equation they represent . Write the correct letter in each box



a	A
b	B
c	C
d	D

	$x + 12 + 38 = 19 + 38$
	$3(x + 12) = 57$
	$x + 12 + 2x + 24 = 19 + 2x + 24$
	$4(x + 12) = 76$